1. Competency #12: Use a multi-step procedure leading to a mathematical solution, conclusion, or generalisation

> *"Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation"*

- Highly relevant: Solving the problem involves several logical steps:
  1. Extract The Kicking Kangaroos' sales data from the chart for February through June;
  2. Observe and confirm the "continuous decline";
  3. Determine the nature of the trend (e.g., approximately linear);
  4. Extrapolate to estimate July's sales.
- This is a classic example of a multi-stage reasoning and calculation process, fully aligned with Competency #12.

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2. Competency #13: Use an understanding of the context to guide or expedite the mathematical solving process (e.g., choosing appropriate precision)

> *"Use an understanding of the context to guide or expedite the mathematical solving process..."*

- Highly relevant: The stem explicitly states that the manager is "worried about declining sales" and asks what happens "if this negative trend continues." This contextual framing guides students to:
  - Avoid overly complex models (e.g., high-degree polynomial fits) and instead adopt a simple, reasonable assumption (e.g., linear decrease);
  - Recognize that the answer should be a whole number within a plausible range (sales are unlikely to jump to 1340 or crash to 70 without extreme circumstances);
  - Understand that "estimate" implies approximation rather than exact calculation.
- Context directly shapes modeling choices and acceptable precision.

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3. Competency #14: Make generalisations based on the results of applying mathematical procedures to find solutions

> *"Make generalisations based on the results of applying mathematical procedures to find solutions"*

- Strongly relevant: Students must generalise a pattern from the specific sequence of sales (Feb–Jun)—for instance, "sales drop by roughly 250 units per month"—and extend this rule to July.
- This embodies the core idea of inductive reasoning in mathematical modeling: moving from observed instances to a predictive rule.

4. Competency #7: Manipulate numbers, graphical and statistical data, algebraic expressions, equations, and geometric representations

> *"Manipulate numbers, graphical and statistical data... and geometric representations"*

- Moderately relevant: Students need to:
  - Accurately read values from the bar chart (e.g., Feb ≈ 1500, Mar ≈ 1250, Apr ≈ 1000, May ≈ 750, Jun ≈ 500);
  - Compute month-to-month differences (≈ –250/month);
  - Estimate July ≈ 500 – 250 = 250 (though the closest option is B. 370, suggesting the trend may not be perfectly linear or requires visual adjustment).
- While numerical and graphical manipulation is involved, the calculations are relatively simple, so relevance is slightly lower than the top three.

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5. Competency #9: Use mathematical tools (including technology, simulation, computational thinking) to help find exact or approximate solutions

> *"Use mathematical tools... to help find exact or approximate solutions"*

- Weakly relevant: The task doesn't require calculators, software, or coding, but it implicitly involves mental math or rough sketching (e.g., drawing a trend line or estimating slope).
- In a more advanced version (e.g., with raw data tables), this competency would be more prominent; here, it plays only a minor role.

1. 第12项：使用多步骤程序得出数学解、结论或一般化结果

> *"Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation"*

- 高度相关：解题需完成多个逻辑步骤：
  1. 从图表中提取 The Kicking Kangaroos 从2月到6月的销量数据；
  2. 观察并确认“持续下降”的趋势；
  3. 判断下降模式（如是否近似线性）；
  4. 基于该模式外推7月的估计值。
- 这是一个典型的多阶段推理与计算过程，完全契合第12项的核心要求。

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2. 第13项：利用对情境的理解来指导或加速数学求解过程（例如采用符合情境的精度）

> *"Use an understanding of the context to guide or expedite the mathematical solving process..."*

- 高度相关：题干明确指出“经理担忧销量下降”，并强调“若相同负面趋势继续”——这提示学生：
  - 不应使用复杂模型（如高阶拟合），而应采用简单、合理的趋势假设（如线性递减）；
  - 答案应为整数且数量级合理（销量不可能突增至1340，也不可能骤降至70，除非极端情况）；
  - “估计”一词暗示不需要精确计算，而需合理近似。
- 情境理解直接引导了建模策略和精度选择。

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3. 第14项：基于应用数学程序所得结果进行归纳

> *"Make generalisations based on the results of applying mathematical procedures to find solutions"*

- 强相关：学生需从2–6月的具体数据序列中归纳出一般趋势规律（如“每月大约减少300张”），再将此规律推广至7月。
- 这正是“从特例到一般化预测”的典型过程，属于归纳推理在数学建模中的应用。

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4. 第7项：操作数字、图形与统计信息、代数表达式、方程及几何表示

> *"Manipulate numbers, graphical and statistical data... and geometric representations"*

- 中等相关：学生需：
  - 从柱状图中准确读取数值（如2月≈1500，3月≈1250，4月≈1000，5月≈750，6月≈500）；
  - 计算相邻月份的差值（约–250/月）；
  - 推算7月 ≈ 500 – 250 = 250（但选项中最接近的是 B. 370，可能趋势非严格线性，或需目测调整）。
- 虽然涉及数据操作，但计算较简单，因此相关性略低于前三项。

5. 第9项：使用数学工具（含技术、模拟、计算思维）辅助求得精确或近似解

> *"Use mathematical tools... to help find exact or approximate solutions"*

- 弱相关：本题未要求使用计算器、软件或编程，但隐含了心算或草稿纸上的简易建模（如画趋势线、估算斜率）。
- 在更复杂的版本中（如提供原始数据表），此项相关性会更高；当前为图表目测，故仅轻微涉及。

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6. 第3项：实施给定策略以确定数学解

> *"Implement a given strategy to determine a mathematical solution"*

- 弱相关：题目未给出具体策略（如“用线性回归”），而是要求学生自主选择合适策略（如线性外推）。
- 因此，更偏向“选择策略”（见 employ.pdf 第2项）而非“实施给定策略”。